MILITARY SPECIFICATION

COMPUTATIONAL SYSTEMS, AIRBORNE

This specification is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 **Scope.** This specification covers all airborne computer resource requirements ranging from embedded microprocessors to dedicated processors to large computational processing systems, including associated software development tools and facilities It covers hardware and software acquired together or separately.

1.2 **Application.** This specification can be used to generate computer resource requirements for inclusion in system specifications, to specify requirements in a prime item development specification for a system which includes computer resources, and to specify requirements for computer resources as a stand-alone end item. Because each application may require a different depth of coverage, use only those sections and paragraphs of this specification which are appropriate.

1.3 **Deviation.** Any projected design for a given application which will result in improvement of system performance, reduced life cycle cost, or reduced development cost through deviation from this specification, or where the requirements of this specification result in compromise in operational capability, shall be brought to the attention of the procuring activity for consideration of change.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to ASD/ENES, Wright-Patterson AFB, OH 45433-6503 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

FSC 7010

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2 APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein

SPECIFICATIONS

MIL-M-38510 Microcircuit, General Specification for

STANDARDS

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

AFR 122-10 Safety Design and Evaluation Criteria for Nuclear Weapon Systems

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence

3. REQUIREMENTS

3.1 Operational system description. The _____ system shall include the following computation components:

3.1.1 Central processor. Separate central processors shall be identical. These processors shall be identified and shall perform the following functions.

Processor Identification	Functions
_____	_____
_____	_____
_____	_____

3.1.2 Dedicated processor. Dedicated processors, including preprocessors, shall be used to perform the following functions _____.

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3.1.3 **Redundant computational system configurations.** Computational systems shall provide on-line redundant/parallel processing for the following critical functions:

_____

_____

_____

3.1.4 **Computer software configuration items.** Operational computer software shall be divided into the following computer software configuration items (CSCIs) _____.

3.2 **Performance requirements**

3.2.1 **System characteristics**

3.2.1.1 **Processor architecture**

3.2.1.1.1 **Word length.** The _____ processor(s) shall have _____ bits of fixed-point accuracy when using single-precision arithmetic.

3.2.1.1.2 **Standard architecture.** The _____ processor(s) shall use _____ standard instruction set architecture.

3.2.1.1.2.1 **Additional features.** The following additional features shall be implemented as part of the _____ architecture in the _____ processor(s). _____.

3.2.1.1.3 **Standard microprocessors.** The _____ embedded microprocessors shall utilize hardware which has been qualified to MIL-M-38510. To minimize support costs, a single microprocessor type shall be used throughout the _____ weapon system(s).

3.2.1.1.4 **Special architecture requirements.** The following architectural requirements shall apply to the _____ processors.

3.2.1.1.4.1 **Instruction repertoire.** The _____ processor shall execute the following instructions:

_____.

3.2.1.1.4.2 **Address capability.** The _____ processor shall directly address a minimum of _____ words of memory. Indirect memory addressing capability shall include a minimum of _____ words using a _____ technique.

3.2.1.1.4.3 **Interrupt structure.** The _____ processor shall provide the capability of processing _____ levels of interrupts. Priority interrupts shall include _____.

3.2.1.2 **Throughput.** The _____ processor shall execute the _____ instruction mix at an average of _____ operations per second.

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3.2.1.3 **Memory requirements**

3.2.1.3.1 **Read/write memory.** The _____ processor shall include _____ words of electrically alterable memory. This memory shall be implemented as _____ memory.

3.2.1.3.1.1 **Memory protection.** At least _____ words of read/write memory shall be capable of being protected from alteration during normal system operation. This protection shall be structured in blocks of _____ words.

3.2.1.3.2 **Read only memory (ROM).** The _____ processor(s) shall include a minimum of _____ words of read only memory.

3.2.1.3.2.1 **Reprogramming capability.** The read only memory (shall/may) be electrically reprogrammable (without/upon) removal from the _____ processor.

3.2.1.3.3 **Mass storage.** The _____ system shall include _____ words of mass storage. This mass storage shall be accessible by the _____ processor(s). This mass storage shall be loadable by _____ while _____. The mass storage shall have an average access time not exceeding _____ and a data transfer rate of at least _____ words per second.

3.2.1.4 **Reserve and growth capacity**

3.2.1.4.1 **Memory reserve.** The _____ computational system shall provide a minimum of _____ percent of total memory delivered for each memory technology as reserve. This total memory shall include mass storage and data insertion devices.

3.2.1.4.2 **Throughput reserve.** At least _____ percent of the throughput shall be available as usable spare during the worst case (most heavily loaded) execution of the operational software for the _____ processor(s).

3.2.1.4.3 **Memory growth provisions.** Each memory in the _____ computational system shall be expandable by _____ percent. This expansion shall be accomplished by the addition of plug-in units that use available space, power, cooling, and memory-address capabilities.

3.2.1.5 **Electrical transient capability**

3.2.1.5.1 **Recognition of transients.** The _____ processor(s) shall include circuitry capable of detecting any condition of electrical power during which the _____ system cannot be operated without shutdown.

3.2.1.5.2 **Performance without shutdown.** The _____ processor(s) shall be capable of operating without shutdown under normal conditions of electrical power as specified in _____. In addition, the processor(s) shall be able to withstand interruptions of electrical power for periods not exceeding _____ seconds without shutdown.

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3.2.1.5.3 **Shutdown and restart.** Conditions of electrical power which require shutdown of the _____ processor(s) shall require storage of _____ so that processing can be restarted without loss of program or data currently being executed.

3.2.1.5.4 **Memory volatility.** All processors which utilize memory technologies which lose critical memory contents upon loss of power shall provide memory retention during periods of system shutdown of up to _____ in duration.

3.2.1.6 **Nuclear circumvention.** Circumvention techniques may be used to improve the hardness of equipment to radiation and EMP effects. The specific requirements for circumvention are identified in the following paragraphs.

3.2.1.6.1 **EMP circumvention.** Circumvention may be used to protect equipment against EMP-induced transient upsets. Circumvention shall not be used to protect against burnout.

a. Transient upset is defined as the generation of erroneous data or the loss of good data as the direct result of a response caused by an EMP-induced transient.

b. Burnout is defined as the irreversible degradation of a component's characteristic due to an EMP-induced transient that produces a circuit malfunction.

3.2.1.6.1.1 **EMP detectors.** EMP detectors shall sense the bulk common-mode current at critical interface connectors exclusive of the power interface connectors. Critical interface connectors shall be determined by analysis. The outputs from each detector shall be "OR'd" to provide a "logic true" EMP circumvention signal. Each EMP circumvention detector shall be set to actuate at _____ percent of the circuit upset threshold level. The upset level shall be defined at each interface connector by analysis.

3.2.1.6.1.2 **EMP interrupt duration.** The EMP circumvention detection signal shall remain "logic true" until _____ seconds after the EMP transients have subsided to _____ percent of the circuit threshold upset level as determined by a lack of an additional circumvention signal.

3.2.1.6.2 **Radiation circumvention.** When nuclear radiation levels are detected that are one order of magnitude below the levels which can cause a transient upset (as determined by analysis), an internal signal shall command the equipment to enter a hold or idle state. Within _____ seconds after the E or P detection signal returns to "logic false," the processor shall be able to resume normal operation.

3.2.1.6.3 **Power supplies.** EMP circumvention shall not be used on power supply circuits. Power supply circuits shall be hardened against burnout and shall protect other circuitry from burnout and transient upset as a result of EMP transients on the power wires.

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3.2.2 System software requirements

3.2.2.1 Higher order language (HOL). The operational software shall be programmed in the _____ higher order language. No less than _____ percent of the object code loaded into the _____ processor(s) shall be the direct result of compilation of assembly language source code.

3.2.2.2 Assembly language use. Assembly language may be used in subsegments of computer software configuration items which are programmed in HOL.

3.2.2.3 Functional performance. The operational software shall perform all computations, data processing, and interfacing functions well enough for the _____ system to fulfill its requirements.

3.2.2.3.1 Executive. The software executive shall control the execution of all operational software by assigning priority to routines on a _____-level scheme. This control scheme shall be such that software can be tested in a deterministic fashion.

3.2.2.3.2 Built-in-test. The built-in-test software shall exercise sufficient processor functions to provide a _____ percent confidence of correct operation. The built-in-test software shall execute in a _____ mode, but shall execute at least _____ times per second.

3.2.2.3.3 Memory erase. The memory-erase software shall erase all classified data in accordance with _____ upon command of _____. This erasure shall be completed within _____ of the command initiation. A positive indication of erasure completion _____ be displayed to the operator.

3.2.2.3.4 Degraded performance capability. The _____ processor(s) software shall automatically sequence through _____ modes of degraded operation when a failure occurs in the processor or an interfacing system.

3.2.2.3.5 Performance of particular functions. The _____ system software shall provide for the following particular functions _____

3.2.3 Hardware characteristics

3.2.3.1 Electrical power. The _____ computational system shall utilize no more than _____ watts of electrical power of characteristics defined by _____.

3.2.3.2 Cooling. The _____ computational system shall utilize _____ cooling.

3.2.3.3 Environment. The _____ computational system shall operate under environmental conditions defined by _____ and shall survive under non-operational conditions defined by _____.

3.2.3.4 Size. The _____ computational system shall not exceed the dimensions defined by _____

3.2.3.5 Weight. The _____ computational system shall not exceed the weight allocated by _____

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3.3 **Reliability.** Computational subsystem reliability requirements shall be compatible with the _____ system reliability allocation.

3.4 **Maintainability**

3.4.1 **Design characteristics.** The _____ system shall be designed for ease of maintenance. Characteristics shall include:

a. Modularization of software into components no larger than _____ source statements.

b. Use of structured design and programming techniques.

c. Use of standard conventions for naming and typing of variables used in the _____ programming language.

d. Fault isolation through use of BITE circuitry.

e. Fault isolation through use of BIT software.

f. Provisions for loading and running maintenance software on the operational computer.

g. Functional partitioning of the system.

h. Other considerations.

3.4.2 **Software development system.** The software development system shall provide a complete comprehensive development and test environment. It shall consist of all tools, facilities, and software used to support the design, coding, documentation, configuration control, test, verification, and modification of operational software. All elements of the software development system shall be identified and managed as configuration items. As a minimum, the following functional capabilities shall be included:

_____

_____

_____

3.4.3 **Pre-flight test software.** Software shall be developed to be loaded in the _____ processor(s) for use during on-ground pre-flight checkout. This software shall provide simulated inputs to, and monitor outputs from, interfacing equipments and perform tests to verify system operability.

3.5 **Safety**

3.5.1 **Nuclear safety.** The _____ system shall be designed using AFR 122-10 as a guide. Any interface with critical nuclear functions shall be _____ to maintain the existing level of nuclear safety.

3.6 **Human engineering.** Computer controls and displays shall be designed in accordance with _____.

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3.7 Interface requirements

3.7.1 **Inputs.** The _____ system shall be designed to accept the following types of inputs:

_____

_____

_____

3.7.2 **Outputs.** The _____ system shall be designed to provide the following types of outputs:

_____

_____

_____

3.7.3 **Program loading.** Computer programs shall be loaded using the _____ memory loader/verifier. In-flight program loading from onboard the aircraft shall be required.

3.7.4 **Data insertion capability.** A _____ device shall be provided for onboard loading of mission-critical data. This device shall as a minimum be capable of storing _____ words of data and transferring them to the _____ processor(s) at a rate of _____ words per second.

4. VERIFICATION

The verifications specified herein shall verify the ability of _____ to meet the requirements of section 3 herein. All verifications shall be the responsibility of the contractor; the Government reserves the right to witness or conduct any verification.

4.1 **Operational system description.** The identification of the system of which the computation components are a part shall be verified by inspection.

4.1.1 **Central processor.** The identification of central processors shall be verified by inspection.

4.1.2 **Dedicated processor.** The identification of dedicated processors shall be verified by inspection.

4.1.3 **Redundant computational system configurations.** Redundant processing for critical functions shall be verified by test.

4.1.4 **Computer software configuration items.** The CSCI structure shall be verified by inspection.

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4.2 Performance verification

4.2.1 System characteristics

4.2.1.1 Processor architecture

4.2.1.1.1 Word length. The word length of each processor shall be verified by inspection.

4.2.1.1.2 Standard architecture. The use of standard architecture shall be verified by _____.

4.2.1.1.2.1 Additional features. The presence of additional architectural features in a processor shall be verified by test.

4.2.1.1.3 Standard microprocessors. Use of MIL-M-38510 processors shall be verified by inspection.

4.2.1.1.4 Special architecture requirements. The implementation of special architectural requirements shall be verified by test.

4.2.1.1.4.1 Instruction repertoire. Instruction repertoire requirements shall be verified by test.

4.2.1.1.4.2 Address capability. Address capability requirements shall be verified by test.

4.2.1.1.4.3 Interrupt structure. Interrupt structure requirements shall be verified by test.

4.2.1.2 Throughput. Throughput shall be verified by test.

4.2.1.3 Memory requirements

4.2.1.3.1 Read/write memory. Implementation of read/write memory shall be verified by test.

4.2.1.3.1.1 Memory protection. Implementation of a protected area of memory storage shall be verified by test.

4.2.1.3.2 Read-only memory (ROM). The presence of a specified size block of read-only memory shall be verified by analysis.

4.2.1.3.2.1 Reprogramming capability. Reprogrammability of the read only memory shall be verified by test.

4.2.1.3.3 Mass storage. Mass storage requirements shall be verified by test.

4.2.1.4 Reserve and growth capacity

4.2.1.4.1 Memory reserve. Reserve memory shall be verified by demonstration.

4.2.1.4.2 Throughput reserve. Throughput reserve shall be verified by test.

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4.2.1.4.3 **Memory growth provisions.** Memory growth capability shall be verified by demonstration.

4.2.1.5 **Electrical transient capability**

4.2.1.5.1 **Recognition of transients.** Recognition of transients shall be verified by test.

4.2.1.5.2 **Performance without shutdown.** The period of operation without shutdown shall be verified by demonstration.

4.2.1.5.3 **Shutdown and restart.** Shutdown and restart capability shall be verified by _____.

4.2.1.5.4 **Memory volatility.** The memory volatility requirement shall be verified by test.

4.2.1.6 **Nuclear circumvention**

4.2.1.6.1 **EMP circumvention**

4.2.1.6.1.1 **EMP detectors.** This requirement shall be verified by _____

4.2.1.6.1.2 **EMP interrupt duration.** The EMP interrupt duration shall be verified by test.

4.2.1.6.2 **Radiation circumvention.** Processor operation after radiation transient conditions shall be verified by test.

4.2.1.6.3 **Power supplies.** Power supply hardening shall be verified by test

4.2.2 **System software requirements**

4.2.2.1 **Higher order language (HOL).** Use of HOL shall be verified by analysis of source and object code listings.

4.2.2.2 **Assembly language use.** Use of assembly language in non-HOL segments of computer software configuration items shall be verified by analysis.

4.2.2.3 **Functional performance.** Software functional performance shall be verified by systems testing. The following function(s) shall be verified and validated (IV & V'd) by an independent contractor.

_____

_____

_____

4.2.2.3.1 **Executive.** The structure of the executive software shall be verified by _____.

4.2.2.3.2 **Built-in-test.** Built-in-test software shall be verified by test

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4.2.2.3.3 **Memory erase.** Memory-erase capability shall be verified by test.

4.2.2.3.4 **Degraded performance capability.** Degraded performance capability shall be verified by test.

4.2.2.3.5 **Performance of identified functions.** Execution of required functions shall be verified by test.

4.2.3 **Hardware characteristics**

4.2.3.1 **Electrical power.** Electrical power utilization shall be verified by test.

4.2.3.2 **Cooling.** The type of cooling required shall be verified by _____.

4.2.3.3 **Environment.** Environmental performance shall be verified by test.

4.2.3.4 **Size.** Compliance with size requirements shall be verified by test.

4.2.3.5 **Weight.** Compliance with weight requirements shall be verified by test.

4.3 **Reliability.** Computer hardware reliability shall be verified by test.

4.4 **Maintainability**

4.4.1 **Design considerations.** Design for maintenance shall be verified by: _____.

4.4.2 **Software development system.** The capabilities of the software development system shall be verified by test.

4.4.3 **Pre-flight test software.** Verification of the pre-flight test software shall be accomplished by test.

4.5 **Safety**

4.5.1 **Nuclear safety.** Verification of nuclear safety shall be accomplished by _____.

4.6 **Human engineering.** Design of processor-human interfaces in accordance with human engineering principles shall be verified by _____.

4.7 **Interface requirements**

4.7.1 **Inputs.** Verification that required input types have been implemented shall be accomplished by analysis.

4.7.2 **Outputs.** The implementation of output types shall be verified by analysis.

4.7.3 **Program loading.** Program loading shall be verified by test.

4.7.4 **Data insertion capability.** Onboard loading capability shall be verified by test.

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5. PACKAGING

5.1 All deliverable items shall be prepared for shipment as directed by the procuring activity.

6. NOTES

6.1 Intended use. The _____ is intended for use in _____.

6.2 Responsible engineering office. The office responsible for development and technical maintenance of this specification is ASD/ENASC, Wright-Patterson AFB OH 45433-6503. Requests for additional information or assistance on this specification can be obtained from Fred H Banks, ASD/ENASC, Wright-Patterson AFB OH 45433-6503, AUTOVON 785-2262, commercial (513) 255-2262. Any information obtained relating to government contracts must be obtained through contracting officers

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APPENDIX

COMPUTATIONAL SYSTEMS, AIRBORNE

HANDBOOK FOR

10. SCOPE

10.1 **Scope.** This appendix provides the information necessary to tailor sections 3, 4, and 6 of the basic specification (MIL-C-87232) for a specific application.

10.2 **Purpose.** This appendix provides information to assist the government procuring activity in the use of MIL-C-87232.

10.3 **Use.** This appendix is designed to assist the project engineer in tailoring MIL-C-87232. The blanks of the basic specification shall be filled in to meet operational needs of the equipment being developed.

10.4 **Format.** Each requirement of section 3 of the basic specification is discussed in section 30 of this appendix. The order, paragraph numbering, paragraph titles, and text of section 3 are repeated exactly in section 30 to facilitate study. To insure that a performance or design requirement from section 3 and its associated verification requirement in section 4 are tailored as a unit, they have been placed next to each other in section 30. The discussion of each requirement is divided into three parts:

a. Rationale - The reason why this requirement exists, the purpose that this requirement advances

b. Guidance - Instructions and principles for setting the values that are left blank in the basic specification, or for omitting the requirement altogether.

c. Lessons learned - Summaries of field experience relevant to this requirement.

10.5 **Responsible engineering office.** The responsible engineering office (REO) for this appendix is ASD/ENASC, Wright-Patterson AFB OH 45433-6503. The individual who has been assigned the responsibility for this handbook is Frederick H. Banks, ASD/ENASC, Wright-Patterson AFB OH 45433-6503, AUTOVON 785-2262, Commercial 255-2262.

20. APPLICABLE DOCUMENTS

20.1 **References.** The documents referenced in this appendix are not intended to be applied contractually. Their primary purpose is to provide background information for the Government engineers responsible for developing the most appropriate performance values (filling in the blanks) for the requirements contained in the specification proper.

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20.2 **Avoidance of tiering.** Should it be determined that the references contained in this appendix are necessary in writing an RFP or building a contract, excessive tiering shall be avoided by calling out only those portions of the reference which have direct applicability. It is a goal of the Department of Defense that the practice of referencing documents in their entirety be eliminated in order to reduce the tiering effect

20.3 **Government documents**

20.3.1 **Specifications and standards**

STANDARDS

Military

MIL-STD-1553	Aircraft Internal Time Division Command/Response Multiplex Data Bus
MIL-STD-1750	Sixteen-Bit Computer Instruction Set Architecture
MIL-STD-1760	Aircraft Store Electrical Interconnection System
DOD-STD-1815	Ada Programming Language
MIL-STD-1862C	NEBULA Instruction Set Architecture (in preparation)
DOD-STD-2167	Defense System Software Development

20.3.2 **Other government publications**

AFR 205-25	Safeguarding the Single Integrated Operational Plan (SIOP)
AFR 800-14/ AFSC Supplement	Management of Computer Resources in Systems (Vol I)
DODD 5000.31	Computer Programming Language Policy

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30. REQUIREMENTS AND VERIFICATIONS

3.1 **Operational system description.** The _____ system shall include the following computation components.

REQUIREMENT RATIONALE (3.1)

This is an introductory paragraph. Computation components will be described in the paragraphs which follow.

REQUIREMENT GUIDANCE

This paragraph should indicate the level at which the computation requirements are being specified. The blank should be filled in with the name of the weapon system, subsystem or computation subsystem for which the overall requirements are being written.

REQUIREMENT LESSONS LEARNED

4.1 **System description.** The identification of the system of which the computation components are a part shall be verified by inspection.

VERIFICATION RATIONALE (4.1)

This verification assures that the computation hardware, software and documentation are consistently identified.

VERIFICATION GUIDANCE

Hardware nameplates and software identification numbers should be checked against specifications and other documentation.

VERIFICATION LESSONS LEARNED

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3.1.1 **Central processor.** Separate central processors shall be identical. These processors shall be identified and shall perform the following functions:

Processor Identification	Functions

REQUIREMENT RATIONALE (3.1.1)

This paragraph identifies, in terms of functional requirements, which processors are shared among subsystems.

REQUIREMENT GUIDANCE

A list of functions should be provided for each identified central processor.

REQUIREMENT LESSONS LEARNED

4.1.1 **Central processor.** The identification of central processors shall be verified by inspection.

VERIFICATION RATIONALE (4.1.1)

This verification assures that the identification is accurate.

VERIFICATION GUIDANCE

Equipment nameplates should be compared with specification identifications.

VERIFICATION LESSONS LEARNED

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3.1.2 Dedicated processor. Dedicated processors, including microprocessors, shall be used to perform the following functions: _____.

REQUIREMENT RATIONALE (3.1.2)

This paragraph identifies, in terms of functional requirements, all processors which are dedicated to a particular subsystem.

REQUIREMENT GUIDANCE

Each processor should be listed separately along with a list of the functions it performs.

REQUIREMENT LESSONS LEARNED

4.1.2 Dedicated processor. The identification of dedicated processors shall be verified by inspection.

VERIFICATION RATIONALE (4.1.2)

This verification assures that all dedicated processors are identified accurately.

VERIFICATION GUIDANCE

The listing of processors can be compared with equipment markings. Functional requirements can be compared with the subsystem description.

VERIFICATION LESSONS LEARNED

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3.1.3 **Redundant computational system configuration.** Computational systems shall provide on-line redundant/parallel processing for the following critical functions:

_____

_____

_____

REQUIREMENT RATIONALE (3.1.3)

For certain functions, such as automatic flight control, or automatic terrain following, redundancy may be the best way to increase system availability and reliability.

REQUIREMENT GUIDANCE

This requirement should be tailored by identifying the functions which require redundant processing.

REQUIREMENT LESSONS LEARNED

4.1.3 **Redundant computational system configurations.** Redundant processing for critical functions shall be verified by test.

VERIFICATION RATIONALE (4.1.3)

This verification is necessary to insure that required redundant processing is provided.

VERIFICATION GUIDANCE

Tests should be prepared to demonstrate both that required redundant processing is accomplished and that the results achieved by each redundant processor is the same. Tests should also demonstrate the capability of the system to reject the results of any one processor when error is deliberately introduced into the results of that processor.

VERIFICATION LESSONS LEARNED

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3.1.4 **Computer software configuration items.** Operational computer software shall be divided into the following computer software configuration items (CSCIs). _____.

REQUIREMENT RATIONALE (3.1.4)

It is necessary to have computer program configuration items in order to establish software baselines and to control changes in the software. This CSCI listing establishes at what level the software is to be controlled.

REQUIREMENT GUIDANCE

It is necessary to decide how many CSCIs to have and how to apportion functions among the CSCIs. As a minimum, it is usually logical to have separate CSCIs for each processor and for different types of software, such as operational and maintenance software. A balance must be struck between too many CSCIs and too few. Too many will result in excessive documentation; too few will not give precise enough control.

REQUIREMENT LESSONS LEARNED

4.1.4 **Computer software configuration items.** The CSCI structure shall be verified by inspection.

VERIFICATION RATIONALE (4.1.4)

This verification is necessary to insure that the CSCI structure as implemented is consistent with that established in the requirements.

VERIFICATION GUIDANCE

The CSCI listing in 3.1.4 should be compared with the CSCIs as programmed and documented.

VERIFICATION LESSONS LEARNED

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3.2 Performance requirements

3.2.1 System characteristics

3.2.1.1 Processor architecture

3.2.1.1.1 Word length. The _____ processor(s) shall have _____ bits of fixed-point accuracy when using single-precision arithmetic

REQUIREMENT RATIONALE (3.2.1.1.1)

The first decision in selecting a processor architecture is in choosing the word length.

REQUIREMENT GUIDANCE

Supply the word length for each processor. Selection of the appropriate word length involves two decisions. The first decision is whether a standard 16-bit architecture is adequate or whether 32-bit or some extended architecture will be needed. The second decision is whether other considerations such as size or cost dictate the use of an 8-bit architecture.

REQUIREMENT LESSONS LEARNED

4.2 Performance verification

4.2.1 System characteristics

4.2.1.1 Processor architecture

4.2.1.1.1 Word length. The word length of each processor shall be verified by inspection.

VERIFICATION RATIONALE (4.2.1.1.1)

Verification is accomplished to ensure that the appropriate processors have been used.

VERIFICATION GUIDANCE

Verification of each processor identification should insure that a processor of the required word length has been used.

VERIFICATION LESSONS LEARNED

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3.2.1.1.2 Standard architecture. The _____ processor(s) shall use _____ standard instruction set architecture.

REQUIREMENT RATIONALE (3.2.1.1.2)

Air Force standardization policy requires use of standard architecture unless there is a compelling rationale for the use of other machines.

REQUIREMENT GUIDANCE

This requirement should be tailored by identifying the standard architecture for each processor listed. MIL-STD-1750 architecture is required for 16-bit application and NEBULA architecture (MIL-STD-1862C, in preparation) is preferred for 32-bit applications. When use of standard architecture is inappropriate, omit this paragraph.

REQUIREMENT LESSONS LEARNED

4.2.1.1.2 Standard architecture. The use of standard architecture shall be verified by _____.

VERIFICATION RATIONALE (4.2.1.1.2)

This verification will insure compliance with Air Force policy.

VERIFICATION GUIDANCE

Compare processor identification with a list of those which conform to standard architectures. If a processor has not been validated as conforming to standards, it should be submitted to the appropriate agency for validation. For further guidance contact the Deputy for Avionics Control, ASD/AX.

VERIFICATION LESSONS LEARNED

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3.2.1.1.2.1 **Additional features.** The following additional features shall be implemented as part of the _____ architecture in the _____ processor(s) _____.

REQUIREMENT RATIONALE (3.2.1.1.2.1)

Standard processor architectures, such as MIL-STD-1750, have some optional features. If these features are needed they should be specified here.

REQUIREMENT GUIDANCE

For each processor, provide the name of the processor, the standard architecture of the processor, and the required optional architectural features. For MIL-STD-1750 architecture, the table of contents of that standard identifies which of the general requirements (section 4) are optional. If no optional features are required, this paragraph may be omitted.

REQUIREMENT LESSONS LEARNED

4.2.1.1.2.1 **Additional features.** The presence of additional architectural features in a processor shall be verified by test.

VERIFICATION RATIONALE (4.2.1.1.2.1)

It is important to verify that all required features are operational.

VERIFICATION GUIDANCE

A program should be executed which exploits each of the required architectural features.

VERIFICATION LESSONS LEARNED

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3.2.1.1.3 **Standard microprocessors.** The _____ embedded microprocessors shall utilize hardware which has been qualified to MIL-M-38510. A single microprocessor type shall be used throughout the _____ weapon system(s).

REQUIREMENT RATIONALE (3.2.1.1.3)

Air Force policy requires the use of microprocessors qualified to MIL-M-38510. It is preferable to use a single type of qualified microprocessor throughout a particular system to minimize support costs.

REQUIREMENT GUIDANCE

This requirement should be tailored by identifying the microprocessor and supplying the weapon system name. A current list of qualified microprocessors can be obtained from RADC. Select one which best fits the system needs in terms of size, power, speed and capacity.

REQUIREMENT LESSONS LEARNED

4.2.1.1.3 **Standard microprocessors.** Use of MIL-M-38510 processors shall be verified by inspection.

VERIFICATION RATIONALE (4.2.1.1.3)

This verification is necessary to insure that appropriate microprocessors are used.

VERIFICATION GUIDANCE

The identified microprocessors should be compared with those on the MIL-qualified list.

VERIFICATION LESSONS LEARNED

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3.2.1.1.4 **Special architecture requirements.** The following architectural requirements shall apply to the _____ processors.

REQUIREMENT RATIONALE (3.2.1.1.4)

Valid technical requirements may exist which cannot be met using standard architectures

REQUIREMENT GUIDANCE

This paragraph should be used only when technical requirements exist which cannot be satisfied by the use of standard architecture processors All critical technical requirements should be listed.

REQUIREMENT LESSONS LEARNED

4.2.1.1.4 **Special architecture requirements.** The implementation of special architectural requirements shall be verified by test.

VERIFICATION RATIONALE (4.2.1.1.4)

This verification is necessary to ensure that all required features are implemented.

VERIFICATION GUIDANCE

A program should be executed which utilizes each of the required features.

VERIFICATION LESSONS LEARNED

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3.2.1.1.4.1 **Instruction repertoire.** The _____ processor shall execute the following instructions: _____.

REQUIREMENT RATIONALE (3.2.1.1.4.1)

This paragraph identifies the required instruction repertoire for processors that use non-standard architecture.

REQUIREMENT GUIDANCE

Identify the instructions in one of the following ways:

a. List them.

b. Refer to an existing documented set of instructions

c. Identify the required special instructions as a minimum set.

REQUIREMENT LESSONS LEARNED

4.2.1.1.4.1 **Instruction repertoire.** Instruction repertoire requirements shall be verified by test.

VERIFICATION RATIONALE (4.2.1.1.4.1)

The implementation of all required instructions can be verified by executing those instructions.

VERIFICATION GUIDANCE

A computer program should be written and executed which uses all required instructions.

VERIFICATION LESSONS LEARNED

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3.2.1.1.4.2 Address capability. The _____ processor shall directly address a minimum of _____ words of memory. Indirect memory addressing capability shall include a minimum of _____ words using a _____ technique.

REQUIREMENT RATIONALE (3.2.1.1.4.2)

This paragraph permits the identification of required memory size

REQUIREMENT GUIDANCE

Specify the number of words in multiples of standard memory module sizes, (32K, 64K, etc.). The technique should be either a paging technique or an indirect addressing technique.

REQUIREMENT LESSONS LEARNED

4.2.1.1.4.2 Address capability. Address capability requirements shall be verified by test.

VERIFICATION RATIONALE (4.2.1.1.4.2)

A test can be devised which verifies the direct and indirect memory addressing capabilities.

VERIFICATION GUIDANCE

A program should be written which accesses both direct and indirect address locations and access times noted to show both modes are operational.

VERIFICATION LESSONS LEARNED

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3.2.1.1.4.3 **Interrupt structure.** The _____ processor shall provide the capability of processing _____ levels of interrupts. Priority interrupts shall include: _____.

REQUIREMENT RATIONALE (3.2.1.1.4.3)

Because of their importance some interrupts must be processed before the functions are executed.

REQUIREMENT GUIDANCE

Processing requirements should be analyzed to determine how many levels of interrupts are necessary.

REQUIREMENT LESSONS LEARNED

4.2.1.1.4.3 **Interrupt structure.** Interrupt structure requirements shall be verified by test.

VERIFICATION RATIONALE (4.2.1.1.4.3)

A test will demonstrate in what order different level interrupts are processed.

VERIFICATION GUIDANCE

A program should be written to demonstrate what happens when interrupts of different priority levels are received and processed.

VERIFICATION LESSONS LEARNED

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3.2.1.2 **Throughput.** The _____ processor shall execute the _____ instruction mix at an average of _____ operations per second.

REQUIREMENT RATIONALE (3.2.1.2)

Some applications may require the specification of a minimum processor execution speed.

REQUIREMENT GUIDANCE

This requirement should be tailored by identifying the instruction mix and execution speed for each processor. The instruction mix can be specified either by referencing a known standard mix or by defining the mix in detail. Standard instruction mixes include DAIS, Gibson, and others. The DAIS mix, used to evaluate MIL-STD-1750 architecture implementations, can be obtained from ASD/ENASF, others from technical literature. Execution speed should be specified in either thousands (KOPS) or millions (MOPS) of operations per second.

REQUIREMENT LESSONS LEARNED

4.2.1.2 **Throughput.** Throughput shall be verified by test.

VERIFICATION RATIONALE (4.2.1.2)

Verification is required to demonstrate that each program achieves the required throughput.

VERIFICATION GUIDANCE

Load each processor with a program containing the identified instruction mix and measure the time it takes the processor to execute the program. Throughput can then be calculated from the number of instructions executed and the time taken to execute them.

VERIFICATION LESSONS LEARNED

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3.2.1.3 **Memory requirements**

3.2.1.3.1 **Read/write memory.** The _____ processor shall include _____ words of electrically alterable memory. This memory shall be implemented as _____ memory.

REQUIREMENT RATIONALE (3.2.1.3.1)

Storage is required for parameters whose values change and data which is received during execution of the computer program

REQUIREMENT GUIDANCE

This requirement should be tailored by identifying each processor, read/write memory size, and type of memory.

REQUIREMENT LESSONS LEARNED

4.2.1.3 **Memory requirements**

4.2.1.3.1 **Read/write memory.** Implementation of read/write memory shall be verified by test.

VERIFICATION RATIONALE (4.2.1.3.1)

A test is required to show that memory contents can be changed.

VERIFICATION GUIDANCE

One way to verify proper operation of read/write memory would be to dump contents of memory before and after execution of a program which alters memory storage.

VERIFICATION LESSONS LEARNED

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3.2.1.3.1.1 **Memory protection.** At least _____ words of read/write memory shall be capable of being protected from alteration during normal system operation. This protection shall be structured in blocks of _____ words.

REQUIREMENT RATIONALE (3.2.1.3.1.1)

Parameters may exist which, if loaded before start of a mission, must remain constant during that mission. This requirement may, in part, be used to protect the critical executive front end of the operational software.

REQUIREMENT GUIDANCE

Identify the size of the block(s) of memory which must be protected from alteration. A software memory protection scheme is required.

REQUIREMENT LESSONS LEARNED

4.2.1.3.1.1 **Memory protection.** Implementation of a protected area of memory storage shall be verified by test.

VERIFICATION RATIONALE (4.2.1.3.1.1)

This verification will insure that the protected area of memory remains unaltered during normal operation.

VERIFICATION GUIDANCE

Conduct a test such as the memory read/write test showing that the contents of the protected area remain unaltered.

VERIFICATION LESSONS LEARNED

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3.2.1.3.2 **Read only memory (ROM).** The _____ processor(s) shall include a minimum of _____ words of read only memory.

REQUIREMENT RATIONALE (3.2.1.3.2)

The availability of ROM provides an economical means of program storage for software which is not subject to frequent change.

REQUIREMENT GUIDANCE

This requirement should be tailored by adding the identification of each processor using ROM and the size of the ROM.

REQUIREMENT LESSONS LEARNED

4.2.1.3.2 **Read only memory (ROM).** The presence of a specified size block of read only memory shall be verified by analysis.

VERIFICATION (4.2.1.3.2)

The requirement for ROM is primarily economic. Nevertheless, it is important to verify that the type of memory requested is present.

VERIFICATION GUIDANCE

ROM can be identified by comparing program listings with the results of the memory read/write tests.

VERIFICATION LESSONS LEARNED

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3.2.1.3.2.1 **Reprogramming capability.** The read only memory (shall/may) be electrically reprogrammable (without/upon) removal from the _____ processor.

REQUIREMENT RATIONALE (3.2.1.3.2.1)

This requirement will permit infrequent alteration of memory contents should requirements change.

REQUIREMENT GUIDANCE

This requirement should be tailored by identifying the processor and whether memory can be reprogrammed with an on-board loader or in a laboratory environment.

REQUIREMENT LESSONS LEARNED

Non-reprogrammable ROM has been used in many systems where requirements were not expected to change. When changes did occur, it was necessary to throw away the memory devices and buy new memory.

4.2.1.3.2.1 **Reprogramming capability.** Reprogrammability of the read only memory shall be verified by test.

VERIFICATION RATIONALE (4.2.1.3.2.1)

It is important to verify that the ROM can be changed. If the memory cannot be changed, it may be necessary to buy new memory if software changes are required.

VERIFICATION GUIDANCE

Contents of the ROM should be changed and new and old values compared.

VERIFICATION LESSONS LEARNED

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3.2.1.3.3 **Mass storage.** The _____ system shall include _____ words of mass storage. This mass storage shall be accessible by the _____ processor(s). This mass storage shall be loadable by _____ while _____. The mass storage shall have an average access time not exceeding _____ and a data transfer rate of at least _____ words per second.

REQUIREMENT RATIONALE (3.2.1.3.3)

Some systems may require mass storage for mission data overlays, terrain maps, maintenance software overlays for operational computers, reprogramming after memory loss, on-line operating system functions, or other requirements.

REQUIREMENT GUIDANCE

This requirement should be tailored by identifying the size and type of storage and the processors which have access to the mass storage. The loading mechanism should be identified as "insertion of disc/insertion of cassettes/ connection to a memory loader-verifier while in flight/on the ground." Loading may also be accomplished from the data insertion device via the processor.

REQUIREMENT LESSONS LEARNED

4.2.1.3.3 **Mass storage.** Mass storage requirements shall be verified by test.

VERIFICATION RATIONALE (4.2.1.3.3)

It is important to demonstrate that the identified processors can load data from different locations on mass storage within the times specified.

VERIFICATION GUIDANCE

Data patterns to be overlaid should be loaded into different locations of the mass storage. Time to load this data into each of the selected processors should be measured and the contents of processor memory verified.

VERIFICATION LESSONS LEARNED

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3.2.1.4 **Reserve and growth capacity**

3.2.1.4.1 **Memory reserve.** The _____ computational system shall provide a minimum of _____ percent of total memory delivered for each memory technology as reserve. This total memory shall include mass storage and data insertion devices.

REQUIREMENT RATIONALE(3.2.1.4.1)

Sufficient memory reserve must be maintained to accommodate future design changes as well as maintenance actions.

REQUIREMENT GUIDANCE

Tailor the requirement by specifying a percentage of total memory at a given point in development for each processor. Nominal memory reserves are usually at least 30 percent, and may be higher, depending on the number of functions serviced, availability of other memory alternatives, type of function(s) and potential for change. In some cases you may need to specify individual reserve requirements for each memory technology. This may necessitate having blanks in the specification to be filled in by each offeror as part of his proposal.

REQUIREMENT LESSONS LEARNED

4.2.1.4 **Reserve and growth capcity**

4.2.1.4.1 **Memory reserve.** Reserve memory shall be verified by demonstration.

VERIFICATION RATIONALE (4.2.1.4.1)

If reserve memory is less than specified, the system may be unable to accommodate either changes or normal maintenance corrections.

VERIFICATION GUIDANCE

Verification can be accomplished by demonstration that the required reserve is available.

VERIFICATION LESSONS LEARNED

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3.2.1.4.2 **Throughput reserve.** At least _____ percent of the throughput shall be available as usable spare during the worst-case (most heavily loaded) execution of the operational software for the _____ processor(s).

REQUIREMENT RATIONALE (3.2.1.4.2)

Software changes usually mean additional time utilization upon execution. Provision needs to be made to accommodate future changes.

REQUIREMENT GUIDANCE

Memory reserve and throughput reserve need to be considered in tandem as a shortage in either may place demands upon the other. It is important to specify timing reserve so that reserve is available in other than background modes of execution and particularly during worst-case timing operations.

REQUIREMENT LESSONS LEARNED

4.2.1.4.2 **Throughput reserve.** Throughput reserve shall be verified by test.

VERIFICATION RATIONALE (4.2.1.4.2)

Only in a special test can the conditions of execution be controlled closely enough to allow accurate measurement of throughput utilization.

VERIFICATION GUIDANCE

It will be necessary to measure time utilization in a sufficient number of modes of operation to ensure that worst-case data is used. Foreground and background (required and not required) processing time will have to be measured separately. Throughput can be calculated from the data. The methods of measurement must be consistent with the conditions of the required specification. A computer program must be implemented to measure throughput in verifying this requirement.

VERIFICATION LESSONS LEARNED

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3.2.1.4.3 Memory growth provisions. Each memory in the _____ computational system shall be expandable by _____ percent. This expansion shall be accomplished by the addition of plug-in units that use available space, power, cooling, and memory address capabilities.

REQUIREMENT RATIONALE (3.2.2.4.3)

Memory expansion provisions allow the addition of memory, should it be needed in the future.

REQUIREMENT GUIDANCE

Tailor this requirement by identifying the processor and the percentage of memory expansion. The percentage of expansion will be determined by the size of the memory modules available for addition, for example, if the only module that can be added is the same as the existing memory, expansion must be 100 percent.

REQUIREMENT LESSONS LEARNED

4.2.1.4.3 Memory growth provisions. Memory growth capability shall be verified by demonstration.

VERIFICATION RATIONALE (4.2.1.4.3)

Only by testing the expanded memory can one be sure that the power supply, addresses, and other provisions are adequate.

VERIFICATION GUIDANCE

The memory should be expanded, loaded, and tested to be sure that power, coding, and addressing are adequate.

VERIFICATION LESSONS LEARNED

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3.2.1.5 **Electrical transient capability**

3.2.1.5.1 **Recognition of transients.** The _____ processor(s) shall include circuitry capable of detecting any condition of electrical power during which the _____ system cannot be operated without shutdown.

REQUIREMENT RATIONALE (3.2.1.5.1)

In order to protect itself against transients, the processor must detect voltage conditions which could damage the circuitry or cause altered logic states.

REQUIREMENT GUIDANCE

This requirement should be tailored by adding the processor and system identification and should be consistent with overall system electrical requirements.

REQUIREMENT LESSONS LEARNED

4.2.1.5 **Electrical transient capability**

4.2.1.5.1 **Recognition of transients.** Recognition of transients shall be verified by test.

VERIFICATION RATIONALE (4.2.1.5.1)

The only way to demonstrate that circuitry can recognize various transients is by actual test.

VERIFICATION GUIDANCE

The processor should be subjected to a variety of power transients of different levels and duration and its ability to detect out-of-tolerance conditions monitored.

VERIFICATION LESSONS LEARNED

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3.2.1.5.2 **Performance without shutdown.** The _____ processor(s) shall be capable of operating without shutdown under normal conditions of electrical power as specified in _____. In addition, the processor(s) shall be able to withstand interruptions of electrical power for periods not exceeding _____ seconds without shutdown.

REQUIREMENT RATIONALE (3.2.1.5.2)

Because of the importance of the processor to system operation, it is required to have the capability to operate on reserve power for short periods of time.

REQUIREMENT GUIDANCE

This requirement should be tailored by identifying the processor, referencing the power specification requirements and determining the period of operation during interruptions of power. This period is a function of the amount of power required and the type of reserve, such as battery or capacitor. For systems that must operate continuously for many hours, you may want to specify a cumulative reserve or period of operation--five minutes during a 12-hour mission, for example.

REQUIREMENT LESSONS LEARNED

4.2.1.5.2 **Performance without shutdown.** The period of operation without shutdown shall be verified by demonstration.

VERIFICATION RATIONALE (4.2.1.5.2)

Operation during periods of interruption can only be demonstrated by test.

VERIFICATION GUIDANCE

A test cycle should be devised which subjects the processor to the longest allowable period of power interruption as well as several shorter periods.

VERIFICATION LESSONS LEARNED

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3.2.1.5.3 **Shutdown and restart.** Conditions of electrical power which require shutdown of the _____ processor(s) shall require storage of _____ so that processing can be restarted without loss of program or data currently being executed.

REQUIREMENT RATIONALE (3.2.1.5.3)

If a computer has to shut down, it is desirable for it to resume processing at the place where it was interrupted, without loss of data or continuity.

REQUIREMENT GUIDANCE

Identify the processor and the parameters such as register contents, memory addresses, and current instructions which must be protected and stored in an identifiable location to facilitate restart. These parameters will vary with the architecture of each machine and the restart scheme used.

REQUIREMENT LESSONS LEARNED

4.2.1.5.3 **Shutdown and restart.** Shutdown and restart capability shall be verified by _____.

VERIFICATION RATIONALE (4.2.1.5.3)

It is necessary to verify shutdown and restart to be sure they function correctly.

VERIFICATION GUIDANCE

If a particular processor has a special instruction which accomplishes the shutdown and restart, it may be possible to verify by analysis. If these operations are done in software, it is preferable to test. If testing is selected, cover as many shutdown contingencies as possible.

VERIFICATION LESSONS LEARNED

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3.2.1.5.4 **Memory volatility.** All processors which utilize memory technologies which lose critical memory contents upon loss of power shall provide memory retention during periods of system shutdown of up to _____ in duration.

REQUIREMENT RATIONALE (3.2.1.5.4)

This requirement is intended to prevent loss of memory when semiconductor memories, which lose contents upon loss of power, are used.

REQUIREMENT GUIDANCE

This requirement should be tailored by identifying the number of hours during which the processor is without system power. For systems with extensive periods of ground time, the period can be quite lengthy.

REQUIREMENT LESSONS LEARNED

4.2.1.5.4 **Memory volatility.** The memory volatility requirement shall be verified by test.

VERIFICATION RATIONALE (4.2.1.5.4)

Testing is required to be certain that the backup power scheme works.

VERIFICATION GUIDANCE

A test should simply shut off system power and then verify that memory contents have been maintained for the period specified.

VERIFICATION LESSONS LEARNED

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3.2.1.6 **Nuclear circumvention.** Circumvention techniques may be used to improve the hardness of equipment to radiation and EMP effects. The specific requirements for circumvention are identified in the following paragraphs.

3.2.1.6.1 **EMP circumvention.** Circumvention may be used to protect equipment against EMP-induced transient upsets. Circumvention shall not be used to protect against burnout.

a. Transient upset is defined as the generation of erroneous data or the loss of good data as the direct result of a response caused by an EMP-induced transient.

b. Burnout is defined as the irreversible degradation of a component's characteristic due to an EMP-induced transient that produces a circuit malfunction.

REQUIREMENT RATIONALE (3.2.1.6.1)

This is not a requirement but an explanation of what circumvention is and how it is to be used.

REQUIREMENT GUIDANCE

No tailoring is necessary. Nuclear circumvention requirements for the airborne computational subsystem must be consistent with the system nuclear and nuclear circumvention requirements. When these requirements are classified it may be useful to incorporate these requirements by reference.

REQUIREMENT LESSONS LEARNED

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3.2.1.6.1.1 **EMP detectors.** EMP detectors shall sense the bulk common mode current at critical interface connectors exclusive of the power interface connectors. Critical interface connectors shall be determined by analysis. The outputs from each detector shall be "OR'd" to provide a "logic true" EMP circumvention signal. Each EMP circumvention detector shall be set to actuate at _____ percent of the circuit upset threshold level. The upset level shall be defined at each interface connector by analysis

REQUIREMENT RATIONALE (3.2.1.6.1.1)

It is necessary to detect the presence of EMP in order to initiate the circumvention mode.

REQUIREMENT GUIDANCE

A circuit and timing analysis may be needed to define the circuit upset threshold level.

REQUIREMENT LESSONS LEARNED

4.2.1.6 **Nuclear circumvention**

4.2.1.6.1 **EMP circumvention**

4.2.1.6.1.1 **EMP detectors.** This requirement shall be verified by _____.

VERIFICATION RATIONALE (4.2.1.6.1.1)

When possible, the capability to detect EMP shall be verified.

VERIFICATION GUIDANCE

Consultation with nuclear certification experts at Kirtland AFB is advised before finalizing this requirement.

VERIFICATION LESSONS LEARNED

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3.2.1.6.1.2 **EMP interrupt duration.** The EMP circumvention detection signal shall remain "logic true" until _____ seconds after the EMP transients have subsided to _____ percent of the circuit threshold upset level as determined by a lack of an additional circumvention signal.

REQUIREMENT RATIONALE (3.2.1.6.1.2)

This requirement is intended to keep the processor in the circumvention mode until EMP transients have subsided.

REQUIREMENT GUIDANCE

Specify the time period needed to ascertain that it is safe to return the processor to service.

REQUIREMENT LESSONS LEARNED

4.2.1.6.1.2 **EMP interrupt duration.** The EMP interrupt duration shall be verified by test.

VERIFICATION RATIONALE (4.2 1.6.1.2)

The duration can readily be tested and measured. It is important to know that the detection circuit will return to a normal state after EMP transients have subsided.

VERIFICATION GUIDANCE

If paragraph 3.2.1.6.1.1 is verified by test, these tests can be combined. If not, the EMP signal can be turned off and the 10-minute delay observed.

VERIFICATION LESSONS LEARNED

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3.2.1.6.2 **Radiation circumvention.** When nuclear radiation levels are detected that are one order of magnitude below the levels which can cause a transient upset (as determined by analysis), an internal signal shall command the equipment to enter a hold or idle state. Within _____ seconds after the E or P detection signal returns to "logic false," the processor shall be able to resume normal operation.

REQUIREMENT RATIONALE (3.2.1.6.2)

This requirement provides for resumption of normal operation after dissipation of transient conditions in addition to EMP conditions previously covered.

REQUIREMENT GUIDANCE

Identify the length of time the processor should remain on hold after transient conditions are no longer detected.

REQUIREMENT LESSONS LEARNED

4.2.1.6.2 **Radiation circumvention.** Processor operation after radiation transient conditions shall be verified by test.

VERIFICATION RATIONALE (4.2.1.6.2)

It is important to know that the processor will properly resume operation after transient upset.

VERIFICATION GUIDANCE

Tests involving nuclear radiation transients should be coordinated with transient upset.

VERIFICATION LESSONS LEARNED

3.2.1.6.3 **Power supplies.** EMP circumvention shall not be used on power supply circuits. These circuits shall be hardened against burnout and shall protect other circuitry from burnout and transient upset as a result of EMP transients on the power wires.

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REQUIREMENT RATIONALE (3.2.1.6.3)

Power supply operation will be required during the shutdown and restart process.

REQUIREMENT GUIDANCE

No tailoring is required.

REQUIREMENT LESSONS LEARNED

4.2.1.6.3 **Power supplies.** Power supply hardening shall be verified by test.

VERIFICATION RATIONALE (4.2.1.6.3)

Proper operation of the hardened circuit can only be verified by test.

VERIFICATION GUIDANCE

A hardness test will require coordination with a nuclear radiation facility.

VERIFICATION LESSONS LEARNED

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3.2.2 **System software requirements**

3.2.2.1 **Higher order language (HOL).** The operational software shall be programmed in the _____ higher order language. No less than _____ percent of the object code loaded into the _____ processor(s) shall be the direct result of compilation of assembly language source code.

REQUIREMENT RATIONALE (3.2.2.1)

Air Force policy (AFR 800-14) requires the use of standard HOL's to write software.

REQUIREMENT GUIDANCE

This requirement should be tailored by selecting the appropriate HOL for each processor. Approved HOL's are identified in DODD 5000.31 and AFSC supplement 1 to AFR 800-14. Assess each processor for functions (such as time-dependent I/O) which cannot be accomplished in HOL and estimate the percentage.

REQUIREMENT LESSONS LEARNED

4.2.2 **System software requirements**

4.2.2.1 **Higher order language (HOL).** Use of HOL shall be verified by analysis of source and object code listings.

VERIFICATION RATIONALE (4.2.2.1)

Source code listings should be adequate to identify the language in which the software is written.

VERIFICATION GUIDANCE

Percentages of non-HOL object code can be computed from percentage of source code and appropriate expansion ratios. These expansion ratios should be computed from actual object code listings.

VERIFICATION LESSONS LEARNED

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3.2.2.2 **Assembly language use.** Assembly language may be used in subsegments of computer software configuration items which are programmed in HOL.

REQUIREMENT RATIONALE (3.2.2.2)

Some portion of the software may not be codable in HOL because of a need to control timing or other parameters. Such software should be written in assembly language since assembly language is more readily maintained than machine language.

REQUIREMENT GUIDANCE

No tailoring is required.

REQUIREMENT LESSONS LEARNED

4.2.2.2 **Assembly language use.** Use of assembly language in non-HOL segments of computer software configuration items shall be verified by analysis.

VERIFICATION RATIONALE (4.2.2.2)

This verification is required to assure that non-HOL segments are written in assembly and not machine language.

VERIFICATION GUIDANCE

Analysis of source code listings should identify each language used.

VERIFICATION LESSONS LEARNED

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3.2.2.3 **Functional performance.** The operational software shall perform all computations, data processing, and interfacing functions well enough for the _____ system to fulfill its requirements.

REQUIREMENT RATIONALE (3.2.2.3)

Software must do whatever is necessary, within the limits of the hardware system, to meet system performance.

REQUIREMENT GUIDANCE

This requirement should be tailored by identifying the system that the airborne computational system supports, or is a part of.

REQUIREMENT LESSONS LEARNED

4.2.2.3 **Functional performance.** Software functional performance shall be verified by systems testing. The following function(s) shall be verified and validated (IV & V'd) by an independent contractor:

_____

_____

_____

VERIFICATION RATIONALE (4.2.2.3)

Software performance is not acceptable until the system as a whole has demonstrated satisfactory operation.

VERIFICATION GUIDANCE

System testing needed for software acceptance should include total system ground and flight tests. Compare the cost of independent verification and validation (IV & V) of each function with the cost of an error in that function. List only those functions where the cost of IV & V is less than the cost of an error.

VERIFICATION LESSONS LEARNED

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3.2.2.3.1 **Executive.** The software executive shall control the execution of all operational software by assigning priority to routines on a _____-level scheme. This control scheme shall be such that software can be tested in a deterministic fashion.

REQUIREMENT RATIONALE (3.2.2.3.1)

This requirement presents the opportunity to specify the number of interrupt priority levels.

REQUIREMENT GUIDANCE

This requirement should be tailored by specifying the number of interrupt priority levels.

REQUIREMENT LESSONS LEARNED

4.2.2.3.1 **Executive.** The structure of the executive software shall be verified by _____.

VERIFICATION RATIONALE (4.2.2.3.1)

This requirement must be verified to assure that the executive is implemented in such a way as to facilitate deterministic operation and test.

VERIFICATION GUIDANCE

You may choose to verify the executive structure by demonstration or test. This decision should be based in part on the complexity of the executive. The more complex the executive, the more perferable testing becomes.

VERIFICATION LESSONS LEARNED

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3.2.2.3.2 **Built-in-test.** The built-in-test software shall exercise sufficient processor functions to provide a _____ percent confidence of correct operation. The built-in-test software shall execute in a _____ mode, but shall execute at least _____ times per second.

REQUIREMENT RATIONALE (3.2.2.3.2)

Built-in-test refers to test of the processor itself. This can be accomplished in part with special software.

REQUIREMENT GUIDANCE

Specify the accuracy and frequency of the built-in-test and whether it should execute in a foreground or background mode.

REQUIREMENT LESSONS LEARNED

4.2.2.3.2 **Built-in-test.** Built-in-test software shall be verified by test.

VERIFICATION RATIONALE (4.2.2.3.2)

A statistical confidence of correct built-in-test operation can only be achieved by testing.

VERIFICATION GUIDANCE

Testing should introduce known faults and verify the built-in-test's ability to identify them.

VERIFICATION LESSONS LEARNED

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3.2.2.3.3 **Memory erase.** The memory-erase software shall erase all classified data in accordance with _____ upon command of _____. This erasure shall be completed within _____ of the command initiation. A positive indication of erasure completion _____ be displayed to the operator.

REQUIREMENT RATIONALE (3.2.2.3.3)

Provisions need to be made to avoid disclosure of classified data should a processor be lost.

REQUIREMENT GUIDANCE

Identify the document that tells what classified data to erase. For an SIOP environment, call out AFR 205-25. In the other blanks, indicate how the memory erase is to be commanded, how fast it is to be completed, and whether or not an indication should be displayed to the operator.

REQUIREMENT LESSONS LEARNED

4.2.2.3.3 **Memory erase.** Memory-erase capability shall be verified by test.

VERIFICATION RATIONALE (4.2.2.3.3)

The adequacy of the memory-erase function can only be verified by test.

VERIFICATION GUIDANCE

Upon completion of a memory-erase cycle an attempt should be made to reconstruct memory contents. Memory erase is adequate if this cannot be done.

VERIFICATION LESSONS LEARNED

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3.2.2.3.4 Degraded performance capability. The _____ processor(s) software shall be implemented so as to automatically sequence through _____ modes of degraded operation upon identification of failure of equipment interfacing with or within the processor subsystem.

REQUIREMENT RATIONALE (3.2.2.3.4)

It may be possible to operate in a less-than-optimum mode after equipment failures.

REQUIREMENT GUIDANCE

This requirement should be tailored by identifying the sequencing provision and the modes of failed operation. These modes can be identified by constructing a matrix of operational modes versus equipment failures.

REQUIREMENT LESSONS LEARNED

4.2.2.3.4 Degraded performance capability. Degraded performance capability shall be verified by test.

VERIFICATION RATIONALE (4.2.2.3.4)

It will be necessary to demonstrate what actually happens when an equipment fails.

VERIFICATION GUIDANCE

Failures should be introduced and the ability of the processor to switch to degraded modes monitored.

VERIFICATION LESSONS LEARNED

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3.2.2.3.5 **Performance of particular functions.** The _____ system software shall provide for the following particular functions. _____.

REQUIREMENT RATIONALE (3.2.2.3.5)

This requirement provides a listing of essential functions.

REQUIREMENT GUIDANCE

Identify the system and list the required functions.

REQUIREMENT LESSONS LEARNED

4.2.2.3.5 **Performance of identified functions.** Execution of required functions shall be verified by test.

VERIFICATION RATIONALE (4.2.2.3.5)

A system test will provide satisfactory evidence of proper operation.

VERIFICATION GUIDANCE

This should be accomplished as part of system testing. Be sure that all functions are executed during test.

VERIFICATION LESSONS LEARNED

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3.2.3 Hardware characteristics

3.2.3.1 Electrical power. The _____ computational system shall utilize no more than _____ watts of electrical power of characteristics defined by _____.

REQUIREMENT RATIONALE (3.2.3.1)

The computational system must utilize power of the type available within the air vehicle. The quantity of power utilized should not be more than that apportioned by the system allocation.

REQUIREMENT GUIDANCE

Identify the power quantity in watts or kilowatts and reference the appropriate paragraph of the governing document, such as the system specification or a MIL-SPEC which defines the power characteristics for each processor.

REQUIREMENT LESSONS LEARNED

4.2.3 Hardware characteristics

4.2.3.1 Electrical power. Electrical power utilization shall be verified by test.

VERIFICATION RATIONALE (4.2.3.1)

Verification will insure that the computational system operates properly on system power and that power in excess of that available is not required.

VERIFICATION GUIDANCE

Processor power tests should, if possible, use instrumentation and equipment designed for other equipment power tests.

VERIFICATION LESSONS LEARNED

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3.2.3.2 **Cooling.** The _____ computational system shall utilize _____ cooling.

REQUIREMENT RATIONALE (3.2.3.2)

Processor cooling requirements must be consistent with provisions of the air vehicle.

REQUIREMENT GUIDANCE

Specify the type of cooling, such as forced air, convection, or other, for each processor. Cooling requirements must be consistent with the cooling type, temperatures, and flow rates allocated in the system requirements.

REQUIREMENT LESSONS LEARNED

4.2.3.2 **Cooling.** The type of cooling required shall be verified by _____.

VERIFICATION RATIONALE (4.2.3.2)

This verification will insure that cooling is consistent with that available.

VERIFICATION GUIDANCE

It may be possible to identify the type of cooling required by inspection of the hardware, otherwise equipment cooling must be demonstrated.

VERIFICATION LESSONS LEARNED

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3.2.3.3 **Environment.** The _____ computational system shall operate under environmental conditions defined by _____ and shall survive under non-operational conditions defined by _____.

REQUIREMENT RATIONALE (3.2.3.3)

The computational system must be capable of operating within the environment of the air vehicle.

REQUIREMENT GUIDANCE

This requirement should be tailored by referencing the appropriate environmental conditions for each computational system. This reference may be to appropriate paragraphs of the system specification, to environmental specifications, or to other documents. Environmental requirements shall be consistent with the system environmental requirement in which the computer is embedded.

REQUIREMENT LESSONS LEARNED

4.2.3.3 **Environment.** Environmental performance shall be verified by test.

VERIFICATION RATIONALE (4.2.3.3)

This verification will insure correct equipment operation in the required environment.

VERIFICATION GUIDANCE

A full spectrum of environmental tests should be run on either the computational system alone or operating in conjunction with other equipment. Environmental stress screening should be considered.

VERIFICATION LESSONS LEARNED

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3.2.3.4 **Size.** The _____ computational system shall not exceed the dimensions defined by _____.

REQUIREMENT RATIONALE (3.2.3.4)

The computational hardware must be capable of fitting within the size envelope offered in the air vehicle.

REQUIREMENT GUIDANCE

Processor size should be specified either in terms of a standard configuration, such as one-half ATR, or by the dimensional constraints of a drawing.

REQUIREMENT LESSONS LEARNED

4.2.3.4 **Size.** Compliance with size requirements shall be verified by measurement.

VERIFICATION RATIONALE (4.2.3.4)

This verification will ensure that the processor will fit into the air vehicle.

VERIFICATION GUIDANCE

Measure the hardware.

VERIFICATION LESSONS LEARNED

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3.2.3.5 **Weight.** The _____ computational system shall not exceed the weight allocated in _____.

REQUIREMENT RATIONALE (3.2.3.5)

Weight and balance are highly critical to performance of the air vehicle and must be carefully allocated to individual equipment items.

REQUIREMENT GUIDANCE

Maximum weight allocated to the processor should be specified. If the processor is expandable, you should probably specify the weight of the expanded configuration.

REQUIREMENT LESSONS LEARNED

4.2.3.5 **Weight.** Compliance with weight requirements shall be verified by test.

VERIFICATION RATIONALE (4.2.3.5)

This verification will ensure that the processor does not exceed the weight allocated.

VERIFICATION GUIDANCE

Weigh the processor hardware in the appropriate configuration.

VERIFICATION LESSONS LEARNED

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3.3 **Reliability.** Computational subsystem reliability requirements shall be compatible with the _____ system reliability allocation.

REQUIREMENT RATIONALE (3.3)

The computational subsystem is important for long-term system operation.

REQUIREMENT GUIDANCE

Call out other documents which contain reliability requirements.

REQUIREMENT LESSONS LEARNED

4.3 **Reliability.** Computer hardware reliability shall be verified by test.

VERIFICATION RATIONALE (4.3)

Performance can be measured by test.

VERIFICATION GUIDANCE

Hardware should be tested as part of a reliability demonstration.

VERIFICATION LESSONS LEARNED

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3.4 Maintainability

3.4.1 Design characteristics. The _____ system shall be designed for ease of maintenance. Characteristics shall include:

a. Modularization of software into components no larger than _____ source statements.

b. Use of structured design and programming techniques.

c. Use of standard conventions for naming and typing of variables used in the _____ programming language.

d. Fault isolation through the use of BITE circuitry.

e. Fault isolation through the use of BIT software.

f. Provisions for loading and running maintenance software on the operational computer.

g. Functional partitioning of the system.

h. Other considerations.

REQUIREMENT RATIONALE (3.4.1)

There are several techniques that can provide ease of maintenance. Unless some of these are specified, maintenance will almost certainly be more difficult and time consuming and therefore more costly.

REQUIREMENT GUIDANCE

This requirement should be tailored by adding the system identification, the software module size, and the programming language to be used DOD-STD-2167 should be consulted for additional guidance; Appendix C may be especially helpful.

REQUIREMENT LESSONS LEARNED

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4.4 **Maintainability**

4.4.1 **Design considerations.** Design for maintenance shall be verified by _____.

VERIFICATION RATIONALE (4.4.1)

Design characteristics which improve ease of maintenance are important to the overall maintainability of the system

VERIFICATION GUIDANCE

Verification that each individual characteristic has been incorporated into the design can be accomplished by either analysis or test. Most software characteristics can be verified by analysis of software documentation while verification of most hardware characteristics will require testing.

VERIFICATION LESSONS LEARNED

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3.4.2 **Software development system.** The software development system shall provide a complete comprehensive development and test environment. It shall consist of all tools, facilities, and software used to support the design, coding, documentation, configuration control, test, verification, and modification of operational software. All elements of the software development system shall be identified and managed as configuration items. As a minimum the following functional capabilities shall be included

_____

_____

_____

REQUIREMENT RATIONALE (3.4.2)

Development tools will be needed to maintain the software.

REQUIREMENT GUIDANCE

This requirement should be tailored by identifying the capabilities of the software development system.

REQUIREMENT LESSONS LEARNED

4.4.2 **Software development system.** The capabilities of the software development system shall be verified by test.

VERIFICATION RATIONALE (4.4.2)

Verification will insure that the software development system functions as intended.

VERIFICATION GUIDANCE

Tests should include the capability to make software modification using the software development system.

VERIFICATION LESSONS LEARNED

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3.4.3 **Pre-flight test software.** Software shall be developed to be loaded in the _____ processor(s) for use during on-ground pre-flight checkout. This software shall provide simulated inputs to, and monitor outputs from, interfacing equipments and perform tests to verify system operability.

REQUIREMENT RATIONALE (3.4.3)

Pre-flight system checkout may be simplified and use of ground support equipment minimized by using special software which can be loaded on the airborne processor(s) and used to check out interfacing systems.

REQUIREMENT GUIDANCE

Identify the processor to be used in running the pre-flight test software.

REQUIREMENT LESSONS LEARNED

4.4.3 **Pre-flight test software.** Verification of the pre-flight test software shall be accomplished by test.

VERIFICATION RATIONALE (4.4.3)

This verification will assure that this software functions as intended.

VERIFICATION GUIDANCE

Tests should include pre-flight system checkout and the ability to identify induced malfunctions.

VERIFICATION LESSONS LEARNED

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3.5 **Safety**

3.5.1 **Nuclear safety.** The _____ system shall be designed using AFR 122-10 as a guide. Any interface with critical nuclear functions shall be _____ to maintain the existing level of nuclear safety.

REQUIREMENT RATIONALE (3.5.1)

Current policy requires certification of systems involving nuclear weapons release.

REQUIREMENT GUIDANCE

Add the system identification and the type of nuclear safety verification required. This verification should be a joint agreement with the nuclear safety organization at Kirtland AFB.

REQUIREMENT LESSONS LEARNED

4.5 **Safety**

4.5.1 **Nuclear safety.** Verification of nuclear safety shall be accomplished by _____.

VERIFICATION RATIONALE (4.5.1)

Verification of nuclear safety is a policy requirement.

VERIFICATION GUIDANCE

Personnel in the appropriate office at Kirtland AFB should be contacted for guidance on verification requirements.

VERIFICATION LESSONS LEARNED

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3.6 **Human engineering.** Computer controls and displays shall be designed in accordance with _____.

REQUIREMENT RATIONALE (3.6)

Processors with human interfaces should be designed according to human factors principles to reduce operator error.

REQUIREMENT GUIDANCE

Consult with ASD/ENECH, who has a MIL-PRIME in this area.

REQUIREMENT LESSONS LEARNED

4.6 **Human engineering.** Design of processor-human interfaces in accordance with human engineering principles shall be verified by _____.

VERIFICATION RATIONALE (4.6)

Interface design should be verified to be sure that controllable facotrs that contribute to operator error are minimized.

VERIFICATION GUIDANCE

Consult with ASD/ENECH.

VERIFICATION LESSONS LEARNED

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3.7 Interface requirements

3.7.1 **Inputs.** The _____ system shall be designed to accept the following types of inputs:

_____

_____

_____

REQUIREMENT RATIONALE (3.7.1)

Input types should be identified.

REQUIREMENT GUIDANCE

Provide a listing of input types. Include standard inputs (MIL-STD-1553 and MIL-STD-1760), high-speed inputs, and discrete inputs as appropriate.

REQUIREMENT LESSONS LEARNED

4.7 Interface requirements

4.7.1 **Inputs.** Verification that required input types have been implemented shall be accomplished by analysis.

VERIFICATION RATIONALE (4.7.1)

This verification indicates only the presence of types.

VERIFICATION GUIDANCE

Drawings should be checked to verify the implementation of required inputs.

VERIFICATION LESSONS LEARNED

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3.7.2 Outputs. The _____ system shall be designed to provide outputs the following types of outputs:

_____

_____

_____

REQUIREMENT RATIONALE (3.7.2)

Output types should be identified.

REQUIREMENT GUIDANCE

Provide a listing of output types. Include standard outputs (MIL-STD-1553, MIL-STD-1760), high-speed outputs, and discrete outputs, as appropriate.

REQUIREMENT LESSONS LEARNED

4.7.2 Outputs. The implementation of output types shall be verified by analysis.

VERIFICATION RATIONALE (4.7.2)

This will verify the existence of the output types.

VERIFICATION GUIDANCE

Drawings should be reviewed to indicate implementation of required outputs.

VERIFICATION LESSONS LEARNED

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3.7.3 **Program loading.** Computer programs shall be loaded using the _____ memory loader/verifier. In-flight program loading from onboard the aircraft shall be required.

REQUIREMENT RATIONALE (3.7.3)

A means is required to enter programs into the processor. If possible, a standard loader/verifier should be used.

REQUIREMENT GUIDANCE

This requirement should be tailored by identifying the loader/verifier and any need for an onboard program or bulk data loading.

REQUIREMENT LESSONS LEARNED

4.7.3 **Program loading.** Program loading shall be verified by test.

VERIFICATION RATIONALE(4.7.3)

It is a simple matter to verify loading by performing the load function.

VERIFICATION GUIDANCE

Load data using the device indicated and verify that the data are subsequently in memory.

VERIFICATION LESSONS LEARNED

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3.7.4 **Data insertion capability.** A _____ device shall be provided for onboard loading of mission-critical data. This device shall, as a minimum, be capable of storing _____ words of data and transferring them to the _____ processor(s) at a rate of _____ words per second.

REQUIREMENT RATIONALE (3.7.4)

This requirement should be used when the quantity of the data to be loaded during a mission, such as TERCOM maps, requires such a device.

REQUIREMENT GUIDANCE

This requirement should be tailored by indicating the type of loading device and the rate and quantity of data to be loaded.

REQUIREMENT LESSONS LEARNED

4.7.4 **Data insertion capability.** Onboard loading capability shall be verified by test.

VERIFICATION RATIONALE (4.7.4)

Test is the simplest way to verify the capability of onboard loading.

VERIFICATION GUIDANCE

A test should include loading and verification of actual datathe aircraft system.

VERIFICATION LESSONS LEARNED

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5. PACKAGING

6. NOTES

6.1 **Intended use.** The _____ is intended for use in _____.

GUIDANCE

In the first blank, fill in the name of the computational system you are specifying. In the next blank, name or describe the system(s) of which it will be a part.

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1 DOCUMENT NUMBER	2 DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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